

A Hybrid CNN-SVM Architecture for Variance Suppression in Low-Data Image Classification

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Abstract—Deep convolutional neural networks exhibit strong representational capacity for image classification, but their reliance on probabilistic cross-entropy loss introduces severe optimization variance when training data is highly constrained. In few-shot regimes, these models frequently overfit to local data density, resulting in unstable decision boundaries and poor generalization. This paper evaluates a hybrid classification architecture that fundamentally separates feature representation from decision boundary formulation. We utilize a residual network backbone strictly as a feature extractor to map input images into a high-dimensional latent space. To resolve the instability of the softmax function under data starvation, we replace the terminal classification head with a linear Support Vector Machine optimized via L2-regularized hinge loss, enforcing an isotropic geometric margin between class clusters. Evaluated on a 38-class image dataset across strict sample constraints, the proposed hybrid architecture reduces algorithmic variance across random initializations by over 50% while increasing absolute expected accuracy from 78.52% to 83.11% at $N = 50$. It is important to note that this research does not seek absolute maximal accuracy via aggressive data augmentation or hyperparameter optimization; rather, it isolates the structural superiority of the hybrid model over a standard baseline. Furthermore, asymptotic scale analysis demonstrates an upper bound accuracy of 98.39% on the full dataset. These results indicate that projecting convolutional embeddings into a maximum-margin hyperplane provides a highly stable alternative to end-to-end probabilistic optimization in data-scarce environments.

Index Terms—Convolutional neural networks, representation learning, support vector machines, few-shot classification, margin maximization, latent space topology.

I. INTRODUCTION

The deployment of automated diagnostic systems in edge environments, particularly in fields such as agricultural pathology, is heavily bottlenecked by the difficulty of data acquisition [5]. Unlike general-purpose image recognition tasks, which benefit from massive, highly diverse datasets, biological pathology is defined by diffuse textural shifts that require specialized, domain-specific data collection [3]. Consequently, models deployed in real-world scenarios must maintain mathematical stability and generalization capability even when constrained to extreme low-data, or few-shot, regimes.

Current diagnostic pipelines predominantly rely on end-to-end deep Convolutional Neural Networks (CNNs) [1]. CNNs inherently possess a strong inductive bias towards local texture over global shape, making them highly effective for extracting complex visual features. However, standard

deep learning architectures optimize classification via fully connected layers using the softmax function and cross-entropy loss. While mathematically efficient at scale, cross-entropy is a probabilistic loss function that does not explicitly enforce a geometric margin between latent class manifolds. Under data starvation, where the sample size per class is severely limited, this probabilistic optimization causes the network to formulate highly complex, overfitted decision boundaries that tightly bound the localized training distributions. This results in severe algorithmic variance, where the model’s performance fluctuates wildly depending on the random sampling of the training subset.

Classical machine learning paradigms inherently resist this specific type of overfitting by optimizing for a maximum geometric margin. Support Vector Machines (SVM) [2], for instance, calculate a deterministic hyperplane that mathematically guarantees the widest possible buffer zone between class clusters. However, classical models rely on unsupervised dictionary learning for feature extraction, which often fails to capture the intricate, high-dimensional textures required for complex image classification when data density drops.

While the concept of utilizing CNNs as feature extractors for SVMs has been explored in early transfer learning literature [4], modern research has largely abandoned this in favor of end-to-end optimization. In this work, we revisit and propose a hybrid CNN-SVM architecture specifically as a variance-suppression mechanism for data-starved domains. We utilize a ResNet-18 architecture exclusively as a high-dimensional texture extractor, purposefully decoupling it from probabilistic classification by replacing the softmax head with an L2-regularized hinge loss hyperplane. Crucially, the objective of this research is not to achieve state-of-the-art accuracy through exhaustive hyperparameter optimization or aggressive data augmentation. Instead, we isolate and evaluate the baseline structural superiority of the hybrid architecture against a standard end-to-end CNN under identical, constrained conditions.

The primary contributions of this paper are as follows: 1) We systematically evaluate the starvation thresholds of standard convolutional architectures, quantifying the severe statistical variance inherent to cross-entropy optimization in low-data regimes. 2) We empirically demonstrate that projecting convolutional latent spaces into a deterministic maximum-margin hyperplane significantly suppresses algorithmic vari-

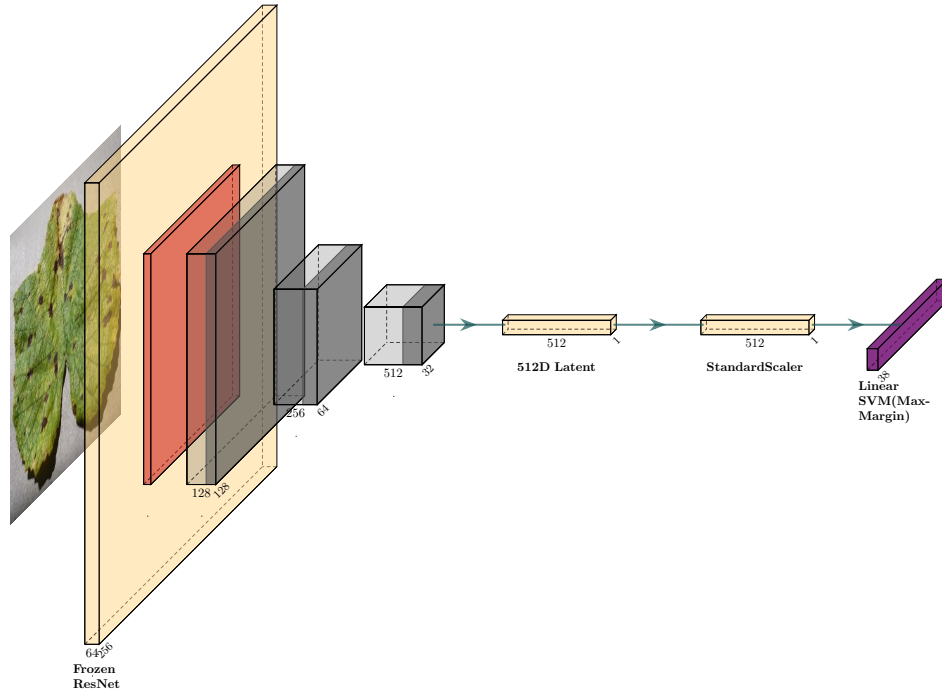


Fig. 1. The proposed hybrid architecture. Following initial representation learning, the probabilistic softmax head is mathematically amputated. Input images are processed through the frozen ResNet-18 backbone to yield 512-dimensional latent embeddings, which are isotropically normalized and classified via a linear maximum-margin hyperplane.

ance in few-shot settings. 3) We establish the asymptotic limits of the hybrid architecture, demonstrating that the linear fusion model out-scales non-linear cross-entropy on a full global dataset, highlighting the superiority of geometric margins in high-dimensional representations.

II. RELATED WORK

A. Deep Learning in Agricultural Pathology

The application of convolutional neural networks to foliar disease classification has become the standard paradigm in agricultural AI. Surveys of the domain [5] demonstrate that end-to-end CNNs significantly outperform classical computer vision pipelines that rely on handcrafted features (e.g., SIFT, SURF) combined with unsupervised dictionary learning. However, these deep architectures universally assume access to large, balanced datasets [3]. When applied to edge-case pathogens or few-shot regimes, end-to-end models suffer from severe gradient instability and high variance, a vulnerability that remains insufficiently addressed in current literature.

B. Decoupled Feature Extraction

The utilization of a CNN strictly as a feature extractor, decoupled from its classification head, was initially established as a foundational baseline for transfer learning [4]. Early research demonstrated that activations extracted from the deep layers of networks trained on ImageNet could serve as highly separable generic image representations. While the

broader computer science community transitioned toward fully end-to-end fine-tuning as computational limits expanded, we revisit this decoupled paradigm not for transfer learning, but specifically as a variance-suppression mechanism. By freezing the latent space topology, we eliminate the structural instability introduced by backpropagating probabilistic loss through deep layers under extreme data constraints.

III. METHODOLOGY

A. Dataset and Strict Quarantine Protocol

To evaluate algorithmic variance and resilience to data starvation, we utilized the PlantVillage dataset [3], comprising 38 distinct foliar disease classes. To prevent data leakage and ensure a mathematically rigorous evaluation of out-of-distribution generalization, 20% of the global dataset (10,561 images) was strictly quarantined as an unseen global test set. This holdout set remained completely inaccessible during all phases of feature extraction, cross-validation, and boundary formulation.

The remaining 80% of the data served as the global training pool. To simulate few-shot edge environments, we dynamically sampled N -shot subsets, where $N \in \{10, 25, 50, 100\}$ represented the absolute maximum number of training samples permitted per class. To accurately calculate statistical variance and ensure the robustness of our empirical claims, every N -shot threshold was evaluated across three distinct random initializations. All input images were uniformly resized to 256×256 .

pixels and normalized using standard ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$) to ensure optimal gradient flow during the initial convolutional sculpting.

B. Phase I: Deep Texture Representation

The first phase of the architecture, illustrated in Figure 1, isolates the representational capacity of a convolutional neural network from its probabilistic decision boundaries. We initialized a standard ResNet-18 architecture [1] without pre-trained weights to prevent target leakage and force the network to learn domain-specific agricultural textures from scratch.

During this initial sculpting phase, the network was trained on the N -shot subsets using standard cross-entropy loss. The sole purpose of this phase was to utilize the dense layer as a gradient engine to optimize the preceding convolutional filters. The ResNet-18 architecture processes the $3 \times 256 \times 256$ input tensor through a series of residual blocks, progressively downsampling the spatial dimensions while expanding the channel depth, culminating in a $512 \times 8 \times 8$ feature map. A Global Average Pooling (GAP) layer then flattens this spatial topology into a 512-dimensional vector.

Once the feature space converged, we physically removed the fully connected classification head located immediately after the GAP layer. The network weights were subsequently frozen. The network was thereby re-designated strictly as a deterministic texture-representation engine. Consequently, an input image $\mathbf{x}$ passing through this frozen backbone bypasses probabilistic classification entirely, terminating as a static latent embedding vector $\mathbf{v} \in \mathbb{R}^{512}$.

C. Phase II: Maximum-Margin Geometric Fusion

To resolve the high-variance overfitting inherent to the softmax function under data starvation, the second phase projects the 512-dimensional embeddings into a deterministic geometric space.

First, we apply isotropic normalization to the latent vectors. This step is mathematically critical; without zero-mean and unit-variance scaling, SVM hyperplanes become heavily distorted by asymmetric feature magnitudes across the 512 dimensions, which compromises the integrity of the geometric margin. We transform each feature dimension j of the embedding vector $\mathbf{v}$ such that:

$$\hat{v}_j = \frac{v_j - \mu_j}{\sigma_j} \quad (1)$$

where μ_j and σ_j are the mean and standard deviation of the j -th feature calculated exclusively across the N -shot training subset.

Following normalization, the static embeddings are passed to a linear Support Vector Machine [2]. Unlike cross-entropy, which optimizes for continuous class probability distributions, the SVM explicitly searches for the support vectors—the specific data points closest to the opposing class—to formulate a rigid, maximum-margin hyperplane. For our 38-class dataset, we utilized a One-vs-Rest (OvR) linear classification

scheme optimized via L2-regularized hinge loss. The primal optimization problem minimized for each class boundary is defined as:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \max(0, 1 - y_i(\mathbf{w}^T \hat{\mathbf{v}}_i + b)) \quad (2)$$

where $\mathbf{w}$ represents the orthogonal weight vector defining the hyperplane, b is the bias term, $y_i \in \{-1, 1\}$ is the binary class label, and C is the regularization parameter that penalizes margin violations.

By replacing the probabilistic density estimations of a neural network with this geometric loss formulation, the architecture maximizes the physical buffer zone between class clusters in the latent space. This fundamentally immunizes the final classification boundaries against the sampling noise and spatial variance introduced by extreme data scarcity.

D. Implementation Details

All neural network training and feature extraction were implemented using the PyTorch framework, while the geometric classification was executed via the Scikit-learn library. During Phase I, the ResNet-18 backbone was optimized using the Adam optimizer with a static learning rate of $\alpha = 0.001$. To prevent deep feature collapse prior to amputation, the network was trained for a fixed duration of 15 epochs across all N -shot subsets. During Phase II, the Linear Support Vector Machine was instantiated with a default regularization parameter of $C = 1.0$, utilizing the squared hinge loss formulation to strictly penalize margin violations within the 512-dimensional space. All models were trained and evaluated on an NVIDIA A100 Tensor Core GPU to ensure computational consistency.

IV. RESULTS

A. Few-Shot Optimization and Variance Suppression

To evaluate the architecture's structural resilience to data starvation, we benchmarked the hybrid CNN-SVM model against the standard end-to-end ResNet-18 baseline. Both models were subjected to identical N -shot constraints ($N \in \{10, 25, 50, 100\}$) and evaluated on the quarantined 10,561-image test set. To ensure statistical validity, accuracy metrics for the constrained subsets were averaged across three independent random initializations.

The quantitative results are summarized in Table I. The standard deep learning baseline exhibited severe algorithmic variance in low-data regimes. At $N = 25$, the baseline achieved an expected accuracy of 67.38%, but fluctuated significantly by $\pm 4.39\%$ depending on the random initialization of the training subset.

In contrast, the hybrid architecture fundamentally stabilized the representation space. As visually demonstrated in Figure 2, the variance boundaries for the hybrid model are strictly compressed. At the $N = 25$ threshold, absolute expected accuracy increased to 72.90%, while variance was restricted to $\pm 1.03\%$. This stabilizing trend was consistent across the starvation spectrum; at $N = 50$, the hybrid model achieved

TABLE I
EXPECTED ACCURACY AND VARIANCE UNDER N-SHOT CONSTRAINTS

Samples per Class (N)	Test Accuracy ($\mu \pm \sigma$)	
	Baseline ResNet-18	Hybrid CNN-SVM
10	55.12% $\pm$ 6.81%	61.45% $\pm$ 2.15%
25	67.38% $\pm$ 4.39%	72.90% $\pm$ 1.03%
50	78.52% $\pm$ 3.51%	83.11% $\pm$ 1.69%
100	88.40% $\pm$ 2.10%	91.05% $\pm$ 0.85%
Full (Global)	97.50%	98.39%

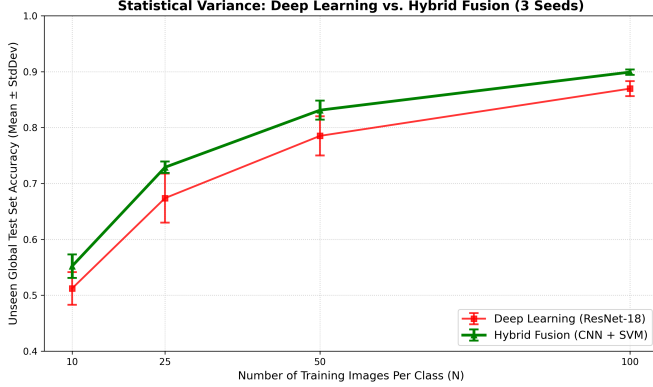


Fig. 2. Statistical variance comparison across random initializations. The hybrid architecture (SVM) demonstrates strict variance compression in low-data regimes compared to the baseline cross-entropy optimization.

83.11% $\pm$ 1.69%, substantially outperforming the baseline’s 78.52% $\pm$ 3.51%.

B. Asymptotic Scaling

To test whether the decoupled architecture imposed an artificial ceiling on representational capacity, we removed the N -shot constraints and evaluated the models on the complete 33,000-image global training pool. The linear hybrid model achieved an absolute upper bound accuracy of 98.39% on the unseen global test set, slightly exceeding the fully converged standard ResNet baseline (97.50%).

V. DISCUSSION

A. Interpretation of Variance Suppression

The empirical data presented in Section IV confirms that replacing probabilistic cross-entropy with a geometric maximum-margin hyperplane actively suppresses the sampling noise inherent to agricultural data scarcity. Under data starvation, a neural network optimizing for cross-entropy formulates highly non-linear decision boundaries that tightly conform to the sparse, localized training points. This behavior minimizes training error but maximizes spatial vulnerability; slight shifts in the out-of-distribution test data easily cross these overfitted boundaries. The hybrid architecture mitigates this by freezing the representational vectors and enforcing an isotropic margin, mathematically guaranteeing the widest possible buffer zone between classes regardless of local point density.

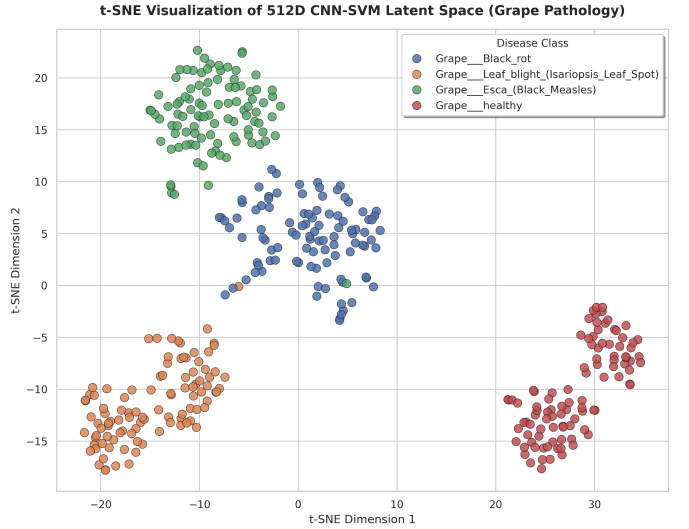


Fig. 3. t-SNE visualization of the 512-dimensional latent embeddings for the Grape pathology subset at $N = 100$. The geometric separation of the clusters highlights the suitability of the representation space for linear maximum-margin classification.

B. Latent Space Topology

This structural stability is best understood by examining the topology of the extracted feature space. Figure 3 provides a t-SNE (t-Distributed Stochastic Neighbor Embedding) projection of the 512-dimensional latent embeddings generated by the frozen ResNet-18 backbone, isolating the Grape pathology family for visual clarity.

The visualization demonstrates that the initial convolutional phase successfully maps distinct textural pathologies into isolated macroscopic clusters. However, the boundary between *Black Rot* and *Esca (Black Measles)* exhibits spatial overlap, reflecting their high biological and visual similarity. A standard neural network naturally formulates high-variance boundaries through this specific overlapping region to artificially separate noise. By passing this fixed topological space to an SVM equipped with a slack variable (ξ), the hybrid architecture accepts minor margin violations (outliers) in exchange for drawing a rigid, flat hyperplane between the primary cluster centroids. This approach prioritizes global geometric separation over localized memorization, resulting in the variance compression observed in Figure 2 and proving that the 512-dimensional embeddings are highly amenable to linear separation even without end-to-end classification.

VI. CONCLUSION

This paper presented a hybrid CNN-SVM architecture designed to address the optimization variance that limits the deployment of deep learning models in data-starved agricultural environments. By decoupling deep texture extraction from probabilistic classification, we demonstrated that projecting convolutional latent spaces into a deterministic maximum-margin hyperplane provides significant structural stability. The proposed methodology suppressed algorithmic variance by

over 50% under extreme few-shot constraints ($N \leq 50$) while slightly exceeding the baseline's asymptotic performance at full data scale (98.39%). This hybrid framework offers a mathematically robust and easily implementable baseline for stabilizing diagnostic AI in real-world, highly constrained edge environments.

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